KMU 330 Test 2 • 2024 Page 1 of 3

KAPASA MAKASA UNIVERSITY

SCHOOL OF APPLIED SCIENCES AND TECHNOLOGY

Department : ICT

Course : Cybersecurity

Year : 3rd Year

20
22

Course Name : Systems Vulnerability Assessment

Course Code : 330

Academic year : 2024

Date : 21st August 2024

Lecturer : Paul M. Mutunda

Task : Test 2

INSTRUCTIONS TO CANDIDATES

1. Write your **Name and Student ID Number** on the Answer sheet provided
2. Marks are shown at the end of each question. Total Marks for this paper is **100%**
3. This paper has **five (5)** questions. **Attempt all**
4. Time allocation is exactly **3 Hours**

CS CamScanner

Question One [20 Marks]

1. List at least **four (4)** benefits of vulnerability analysis [4]
2. Explain in detail, the following types of assessments
    - (i) Application Assessment [2]
    - (ii) Host Assessment [2]
    - (iii) Database Assessment [2]
    - (iv) Network Assessment [2]
3. Differentiate between Risk analysis and vulnerability analysis [2]
4. List and explain any **three (3)** vulnerability assessment tools that you could use in an organisation [6]

Question Two [20 Marks]

1. Discuss the **five (5)** different vulnerability technology [10]
2. Outline the differences between Agent and Agentless [2]
3. List at least **two (2)** advantages and disadvantages of using an Agent [4]
4. Mention **two (2)** benefits and demerits of using an Agentless [4]

– Attack vector
– is an avenue
– privileges to
– Attack
compliance

Question Three [20 Marks]

1. Define a vulnerability Management Lifecyle [2]
2. With the aid of a diagram, explain in detail, the Vulnerability Management Lifecycle [12]
3. Outline **three (3)** clear major reasons of a vulnerability Management Lifecycle [3]
4. Discuss the importance of air – gapping in relation to vulnerability management [3]

Characteristi

Question Four [20 Marks]

1. What is an Attack vector (AV)? [1]
2. Explain in detail, the following metrics under CVSS;
    - (i) Base metrics [5]
    - (ii) Temporal metrics [5]
    - (iii) Environmental metrics [5]
3. According to the vulnerability rating scale under CVSS, when is a software vulnerability considered;
    - (i) Critical 9.0 — 10.0 [1]
    - (ii) High 7.0 — 8.9 [1]
    - (iii) Medium 4.0 — 6.9 [1]
    - (iv) Low 0.0 — 3.9 [1]

exploitability matrica
impact matrics
Exploitability

Severity

access of

Reflects the characteristics of

vulnerability Management life
the process of ~~assess~~, identify, assess, prioritizing and
Remediating vulnerability

200 + 440 − 200
= 440
+ 220
660

673
+ 30
703

660
+ 13
673

Question Five [20 Marks]

1. Give differences between Passive vulnerability analysis and Active vulnerability analysis [2]
2. Outline **five (5)** differences between internal and External passive scanners [10]
3. Give **two (2)** advantages of passive vulnerability analysis [2]
4. Mention **two (2)** disadvantages of passive vulnerability analysis [2]
5. What are the **two (2)** advantages of using an Active vulnerability analysis? [2]
6. List **two (2)** disadvantages of active vulnerability analysis [2]

~~Ac~~passive VA

A'

uso-PL

R'

SAREDS

- Scalabili
- Deploymen complexit
- Efficience
- Resource allocation
- Accuracy
- Security vulnerabi

efficience

Cop
corporate